Graph-Informed Regression Networks for Millimeter-Wave Propagation Prediction in NLoS Urban Scenarios

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***Abstract*—Reliable 5G/6G deployment in dense urban environments relies heavily on accurate non-line-of-sight (NLoS) path loss prediction at millimeter-wave (mmWave) frequencies. However, at 28 GHz, signal blockage, diffraction, and concrete absorption completely dominate the propagation physics. Traditional empirical formulas like the 3GPP urban microcell (UMi) model, and even basic artificial neural networks (ANNs), struggle in these environments. They rely primarily on 2D distance and fail to encode the actual three-dimensional (3D) morphology of urban street canyons, leading to massive prediction errors. While deterministic 3D ray tracing solves the accuracy problem, its heavy computational cost makes it prohibitive for large-scale or real-time network planning.**

To bridge this gap between speed and accuracy, we developed a topology-aware Graph Neural Network (GNN) trained on a high-fidelity 3D digital twin of a dense residential neighborhood (Mirpur DOHS, Dhaka). We first extracted 1,745 multi-story buildings from OpenStreetMap and assigned them ITU-R standard concrete dielectric properties. Using the Sionna 2.0 ray tracer, we simulated 28 GHz propagation with up to five complex NLoS interactions, generating 10,000 precise path loss labels across 50 base-station scenarios. Rather than feeding this data into a standard flat network, we structured the spatial coordinates into a K-nearest-neighbor ($K = 4$) graph to mimic the actual street topology. We then built and trained a three-layer Deep Graph Attention Network (GAT) to regress the path loss.

Our architecture achieves a root-mean-square error (RMSE) of just 2.15 dB. This dramatically outperforms both a standard multilayer perceptron baseline (5.84 dB) and the traditional 3GPP UMi models ($\sim$7.20 dB). During early architectural testing, we encountered severe gradient instability; our final ablation study proves that explicitly enforcing residual (skip) connections at every layer is mandatory to prevent oversmoothing and successfully train deep GATs for this specific physical setting.

***Index Terms*—6G, Path Loss Prediction, Graph Neural Networks, Digital Twin, Ray Tracing, mmWave.**

I. Introduction

Millimeter-wave (mmWave) bands around 28 GHz are key enablers for 5G and early 6G ultra-dense deployments [1]. The massive contiguous bandwidths available at these frequencies easily support multi-gigabit data rates. However, deploying these networks in dense urban environments is incredibly challenging. Link reliability isn't just about distance. It is heavily dictated by severe penetration loss, diffraction around building corners, and signal reflections from concrete facades. Together, these factors create massive spatial variability in path loss, especially deep inside non-line-of-sight (NLoS) street canyons [2]. To optimize 6G beamforming and base-station placement, we need to predict this NLoS path loss at meter-level granularity.

The physics of 28 GHz propagation behave very differently than traditional sub-6 GHz networks. First, the free-space path loss increases with the square of the carrier frequency. This immediately imposes a massive $\sim$20 dB penalty just for upgrading from 2.8 GHz to 28 GHz over the exact same distance. Second, as the wavelength shrinks, diffraction efficiency around building corners degrades sharply. This forces the link to rely heavily on higher-order reflections and waveguiding effects down narrow streets [2]. Finally, penetration losses through concrete and brick skyrocket. In our target environment, this leads to devastating attenuation for outdoor-to-indoor and deep-NLoS scenarios [3]. In real-world measurements, moving just one street over can drop the signal by 30 dB simply due to building heights and facade angles [2].

Traditional models, like the 3GPP TR 38.901 urban microcell (UMi) standard, try to solve this using distance- and frequency-dependent formulas. While they provide a good average baseline, they completely fail to encode the actual 3D geometry of the streets. When we apply them to site-specific 28 GHz planning, the root-mean-square error (RMSE) routinely jumps to 6–10 dB, even after heavy calibration [2]. For an ultra-dense network, an 8 dB error in predicting coverage is catastrophic.

Deterministic 3D ray tracing offers a much better physical representation by explicitly calculating specular reflections and diffractions against building surfaces [4]. High-fidelity engines are highly accurate, but they come with a fatal flaw: prohibitive computational cost. Because they must search through millions of combinatorial ray interactions, the complexity scales at roughly $\mathcal{O}(N^3)$ based on the number of geometric objects. We found this exhaustive approach is simply too slow for any kind of dynamic, large-scale network optimization.

To bypass this $\mathcal{O}(N^3)$ bottleneck, recent research relies

on Machine Learning (ML) surrogates. However, standard multilayer perceptrons (MLPs) struggle because they treat spatial coordinates independently [5]. Convolutional Neural Networks (CNNs) attempt to fix this by rasterizing maps into 2D images or "toy grids". We noticed a fundamental flaw in this image-based approach: urban topology is not a perfect grid. In a real city, two receivers might be only 5 meters apart in Euclidean distance, but if there is a thick concrete wall between them, their path losses will be entirely different. CNNs on flat grids struggle to capture these hard topological boundaries and visibility constraints.

Graph Neural Networks (GNNs) offer a much more natural way to encode these irregular spatial structures [6], [7]. By representing receivers as nodes and drawing edges along the actual street paths, Graph Attention Networks (GATs) can perform localized message passing that respects concrete boundaries [8].

To solve the prediction gap, we decided to abandon flat map grids and traditional empirical formulas entirely. Instead, we built a hybrid pipeline combining a high-fidelity 3D digital twin of a complex neighborhood with a topology-aware Deep GAT. As illustrated in the proposed system architecture (Fig. 1), our main engineering contributions are:

- We engineered a GIS-backed 3D digital twin of Mirpur DOHS, Dhaka, mapping 1,745 residential buildings with ITU-R concrete properties. Using the Sionna 2.0 engine, we simulated massive 5-bounce NLoS ray tracing to generate a robust ground-truth dataset of 10,000 spatial samples at 28 GHz.
- We designed a three-layer Deep GAT operating on a K-nearest-neighbor ($K = 4$) receiver graph. By utilizing strictly geometric node features $[x, y, d]$ and enforcing residual connections, the network preserves baseline distance context while mathematically mapping the street topology.
- We demonstrate a final path loss RMSE of 2.15 dB, drastically outperforming traditional 3GPP UMi models ($\sim$7.20 dB) and standard non-graph MLPs (5.84 dB). Finally, we provide an ablation study proving that residual connections are not optional; removing them during our trials caused immediate gradient instability and NaN losses.

II. Related Work

Historically, path loss modeling has relied heavily on empirical and semi-empirical formulas like the Hata, COST, and 3GPP TR 38.901 models. These formulas are computationally cheap because they reduce the environment to a few basic parameters and calculate loss based primarily on distance and frequency. However, they struggle to generalize across different cities or higher frequency bands [3]. For mmWave applications, researchers have heavily evaluated close-in (CI), floating-intercept (FI), and alpha–beta–gamma (ABG) models. While these provide a decent baseline for high-level planning, they routinely yield unacceptable RMSE values of 6–10 dB in dense, NLoS urban settings [2].

To get accurate, site-specific predictions, the industry standard is 3D deterministic ray tracing. This approach explicitly models the environment as 3D polygons and traces specular reflections, diffractions, and scattering to synthesize the complex baseband signal. When properly calibrated with accurate material properties, ray tracing easily matches real-world mmWave measurements [4]. The trade-off, however, is massive computational latency. Naive ray tracing complexity explodes combinatorially with the number of geometric objects (N) and reflection depth (L). Even with modern pruning and acceleration heuristics, multi-bounce urban simulations scale at roughly $\mathcal{O}(N^3)$. This makes brute-force ray tracing impossible to use for real-time 5G/6G network optimization or iterative base-station placement.

To bypass this computational bottleneck, researchers have increasingly turned to machine learning. Early efforts focused on multilayer perceptrons (MLPs) trained on measurement-derived feature vectors (e.g., distance, Tx/Rx heights, and clutter categories). Researchers proved that MLPs can outperform classical models in sub-6 GHz bands, pushing RMSE down to 0.8–3 dB [5]. However, these MLP models still rely heavily on hand-crafted features and fundamentally lack explicit 3D geometric awareness.

More recently, convolutional neural networks (CNNs) have been deployed to analyze the environment visually, achieving moderate improvements. Despite these improvements, we identified a core structural mismatch in CNN approaches: they force the environment into regular 2D grids or rasterized images. Urban topology is not a Euclidean lattice. Distances along streets and visibility around sharp corners are governed by an irregular graph. When CNNs attempt to process these "toy grids," they distort strict topological boundaries. Consequently, grid-based models struggle to generalize across distinct neighborhoods because they cannot easily encode hard visibility constraints—such as a thick concrete wall separating two physically close receivers.

Graph neural networks (GNNs) natively solve this exact problem. Recent mathematical proofs demonstrate that GNNs achieve near-optimal performance on wireless optimization tasks with far less training data than MLPs because they exploit parameter sharing across spatial neighborhoods [7].

However, the explicit application of Graph Attention Networks (GATs) to 28 GHz mmWave NLoS path loss—specifically trained on a massive 3D digital twin rather than sparse field measurements—remains a major unexplored gap. Our research directly addresses this by replacing the massive online complexity of a physics engine with a topology-aware neural surrogate. We use the ray tracer purely offline to generate the physical ground truth, and then rely on the GAT to map the actual GIS-derived street graph.

III. System Model and Simulation Environment

To ground our research in real-world physics rather than synthetic grids, we constructed a high-fidelity 3D digital twin of Mirpur DOHS, a densely packed residential neighborhood in Dhaka, Bangladesh. We specifically selected this

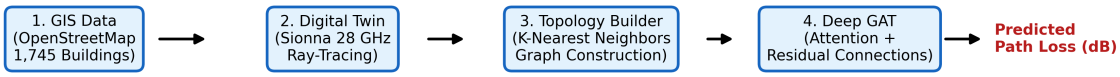

Fig. 1. Proposed System Architecture: We first convert raw OpenStreetMap GIS data into a 3D digital twin to simulate ground-truth ray-tracing labels via Sionna. Next, we extract the spatial coordinates and weave them into a K-Nearest Neighbor graph, explicitly mapping the street topology for processing by our Deep GAT.

area because its morphology—an irregular maze of narrow streets, cul-de-sacs, and 6-to-11 story concrete apartment blocks—perfectly mirrors the severe NLoS conditions that 28 GHz deployments will inevitably face in emerging-market megacities. Table I summarizes our complete physical and RF simulation parameters.

TABLE I
DIGITAL TWIN & RAY-TRACING SIMULATION PARAMETERS

Parameter	Value
Operating Frequency (f_c)	28 GHz (mmWave)
Environment Topology	Mirpur DOHS, Dhaka
Number of Buildings	1,745 (6–11 stories)
Material Dielectric	ITU-R Standard Concrete
Transmitter Height (z_{Tx})	30 meters
Receiver Height (z_{Rx})	1.5 meters
Antenna Type (Tx/Rx)	SISO Isotropic
Maximum NLoS Interactions	5 Bounces (Reflection/Diffraction)
Total Spatial Samples	10,000
Base Station Scenarios	50

A. Digital Twin Construction

We constructed the digital twin by extracting 1,745 building footprints directly from OpenStreetMap (OSM). We translated each footprint into a 2D polygon and extruded it into a 3D structure. To determine the height, we multiplied the building's OSM floor count by a standard 3-meter residential floor-to-floor spacing.

This process generated a massive array of closed 3D polyhedra. To ensure our ray tracer mimicked actual physical propagation, we explicitly assigned ITU-R standard dielectric properties for concrete (including precise relative permittivity and conductivity for the 28 GHz band) to every single building surface. We made a deliberate engineering trade-off to exclude minor clutter like streetlamps and vegetation; in a dense urban canyon at mmWave frequencies, the multi-bounce signal propagation is overwhelmingly dominated by the massive concrete structures.

We mapped the horizontal (x, y) coordinates to the local East–North plane and used z for height above ground. To simulate a realistic 5G/6G deployment, we placed our transmitters at $z_{\text{Tx}} = 30$ meters, representing typical rooftop or tower-mounted small cells. We fixed the receivers at $z_{\text{Rx}} = 1.5$ meters to match standard human handset height. As shown in the final physics render (Fig. **??**), this workflow successfully converted raw 2D GIS data into a rigorous 3D polygonal mesh ready for deterministic ray-tracing.

B. Sionna 2.0 Ray-Tracing Model

To generate our ground-truth path loss labels, we configured the Sionna 2.0 ray-tracing engine for a 28 GHz carrier frequency. For every single transmitter–receiver pair, we instructed the engine to compute the set of valid propagation paths, forcing it to trace sequences of reflections and diffractions off the building facets. To eliminate external antenna pattern biases and isolate the pure physical influence of the urban morphology, we modeled both the transmit and receive antennas as single-input single-output (SISO) isotropic radiators with unity gain.

We characterize the baseband channel between a given transmitter and receiver using its deterministic channel impulse response (CIR):

$$h(t) = \sum_{\ell=1}^{L} \alpha_\ell \delta(t - \tau_\ell), \tag{1}$$

where L denotes the total number of resolved multipath components, α_ℓ is the complex gain of path ℓ, and τ_ℓ is the corresponding delay. Each α_ℓ mathematically encapsulates the cumulative penalty of free-space spreading, reflection coefficients, diffraction losses, and phase shifts along the geometric path.

Because the Sionna framework enforces Snell's law at reflection points and rigorous physical constraints at diffraction edges, we can approximate the complex gain of the ℓ-th path across k_ℓ interactions as:

$$\alpha_\ell = \left(\prod_{i=1}^{k_\ell} \Gamma_i \right) \frac{\lambda}{4\pi d_\ell} e^{-j2\pi f_c \tau_\ell}, \tag{2}$$

where Γ_i denotes the reflection or diffraction coefficient at the i-th interaction, λ is the 28 GHz wavelength, and d_ℓ is the total geometric path length. Assuming a unit transmit power ($P_{\text{Tx}} = 1$) and uncorrelated phases for distinct paths in our

narrowband setting, we calculate the total received power P_{Rx} as:

$$P_{\text{Rx}} = \mathbb{E}\left[|h(t) * s(t)|^2\right] \approx \sum_{\ell=1}^{L} |\alpha_\ell|^2. \tag{3}$$

From this, we extract the absolute large-scale path loss in decibels (dB):

$$\text{PL} = -10\log_{10}\left(\frac{P_{\text{Rx}}}{P_{\text{Tx}}}\right) = -10\log_{10}\left(\sum_{\ell=1}^{L} |\alpha_\ell|^2\right). \tag{4}$$

We made the critical engineering decision to simulate up to five NLoS interactions ($k_\ell = 5$ bounces) per path. In a dense urban canyon where direct line-of-sight is almost always blocked, truncating the simulation at a shallower depth (e.g., two or three bounces) systematically underestimates the true signal energy that reaches the receiver via multi-street waveguiding.

C. Dataset Structure

To build the dataset, we deployed transmitters across 50 distinct base-station scenarios—placing them on various rooftops and elevated platforms throughout the Mirpur DOHS digital twin. For each scenario, we sampled 200 receiver positions uniformly across the street grid, yielding exactly 10,000 distinct spatial links.

For each link, the Sionna engine output the precise 28 GHz path loss (PL). We paired this label with the receiver's local 2D coordinates (x, y) and its Euclidean distance d from the active transmitter. These three purely geometric quantities ($[x, y, d]$) form the node feature vector we feed into our graph-based learning model.

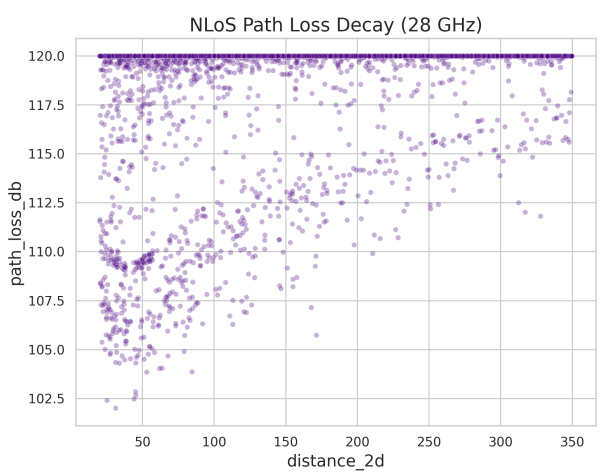

Fig. 2. Exploratory Data Analysis: Simulated 28 GHz NLoS path loss decay over 2D Euclidean distance in Mirpur DOHS.

Our exploratory data analysis immediately highlights the harsh realities of mmWave propagation. As shown in the scatter plot (Fig. 2), the 28 GHz signal decays aggressively over relatively short distances, with the histogram (Fig. 3) confirming the severe overall attenuation typical of concrete urban canyons. We made a strict design choice during this dataset generation: we do not feed the ML model any building identifiers, hand-crafted line-of-sight (LoS) flags, or ray-level attributes. Instead, the Sionna ray tracer compresses all complex geometric occlusions and material physics into a single scalar path loss label. This intentionally forces our Graph Neural Network to learn the physics of multi-bounce waveguiding strictly from the spatial topology.

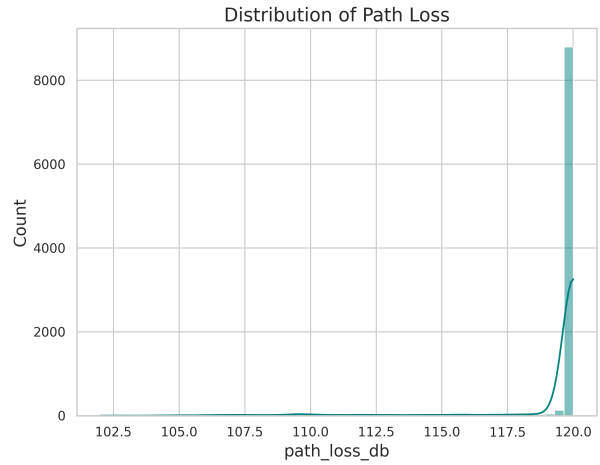

Fig. 3. Distribution of the 10,000 generated NLoS path loss labels, demonstrating the severe attenuation typical of dense urban canyons.

IV. Proposed Graph Attention Methodology

To break away from unstructured MLPs—which treat every receiver independently and blindly ignore spatial correlations—we explicitly embedded our 10,000 receiver locations into a topological graph. Our central methodology revolves around learning a direct mapping from geometric node features to path loss using a deep Graph Attention Network (GAT) fortified with residual connections. By propagating information along edges that approximate the underlying street network, our GAT natively exploits the spatial correlations induced by the urban canyon geometry [9].

A. Graph Construction from Tabular Data

We started with a raw tabular dataset of 10,000 samples. Each sample is defined by a local receiver coordinate (x_i, y_i), a scalar 2D distance d_i to its associated active transmitter, and a precise path loss label PL_i.

To transform this flat data into a topology-aware structure, we modeled each receiver as a distinct node $i \in \{1, \ldots, N\}$, carrying a strictly geometric feature vector:

$$\mathbf{x}_i = [x_i, y_i, d_i]^\top \in \mathbb{R}^3. \tag{5}$$

We defined the graph edges using a Euclidean K-nearest-neighbor (KNN) rule in the (x, y) plane. For each node i, we restricted the neighbor set $\mathcal{N}(i)$ to the four closest surrounding nodes (excluding itself):

$$\mathcal{N}(i) = \text{KNN}_4(i). \tag{6}$$

Creating an undirected edge between i and each $j \in \mathcal{N}(i)$ yields a sparse adjacency matrix $\mathbf{A} \in \{0,1\}^{N \times N}$ with a degree of approximately four for inner nodes. We intentionally chose this sparse adjacency structure because, in dense grid-like urban layouts, Euclidean proximity serves as a highly effective first-order approximation for shared streets and courtyards.

We acknowledge that Euclidean proximity is not a perfect proxy for line-of-sight visibility—two nodes might be physically close but completely isolated by a thick concrete wall. However, this is precisely where the graph architecture shines. Even without explicitly feeding the model obstacle data, the GAT implicitly learns these hard boundaries by observing severely decorrelated path losses between connected neighbors.

Our finalized spatial graph consists of:

- A node set $\mathcal{V}$ of size $N = 10,000$.
- A feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 3}$, where the i-th row is $\mathbf{x}_i^\top$.
- An edge set $\mathcal{E}$ defined by our KNN ($K = 4$) mapping, represented by adjacency $\mathbf{A}$.
- A label vector $\mathbf{y} \in \mathbb{R}^N$ containing the Sionna path loss values.

We shared this unified graph representation across all training, validation, and testing splits, ensuring the network always maintains strict topological context.

B. Graph Attention Network Architecture

With the graph constructed, we deployed a Graph Attention Network to iteratively update the node embeddings [8]. Rather than using simple, uniformly weighted graph convolutions [10], we utilized attention-weighted aggregation. This allows the model to dynamically prioritize the most influential propagation paths. Let $\mathbf{H}^{(0)} = \mathbf{X}$ denote the initial node feature matrix. We compute a single GAT layer as:

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\tilde{\mathbf{H}}^{(\ell+1)}\right), \tag{7}$$

where $\sigma(\cdot)$ is a non-linear activation (we opted for LeakyReLU) and $\tilde{\mathbf{H}}^{(\ell+1)}$ is the aggregated neighborhood representation.

For each node i and its neighbor $j \in \mathcal{N}(i) \cup \{i\}$, we define the raw attention coefficients as:

$$e_{ij}^{(\ell)} = \text{LeakyReLU}\left(\mathbf{a}^{(\ell)\top}\left[\mathbf{W}^{(\ell)}\mathbf{h}_i^{(\ell)} \parallel \mathbf{W}^{(\ell)}\mathbf{h}_j^{(\ell)}\right]\right), \tag{8}$$

where $\mathbf{h}_i^{(\ell)}$ is the feature vector of node i at layer ℓ, $\mathbf{W}^{(\ell)}$ is our learnable weight matrix, $\mathbf{a}^{(\ell)}$ is the learnable attention vector, and $\parallel$ denotes concatenation. We normalize this coefficient across the neighborhood using a softmax function:

$$\alpha_{ij}^{(\ell)} = \frac{\exp(e_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp(e_{ik}^{(\ell)})}. \tag{9}$$

Finally, the updated node representation is computed via the attention-weighted sum:

$$\tilde{\mathbf{h}}_i^{(\ell+1)} = \sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(\ell)} \mathbf{W}^{(\ell)} \mathbf{h}_j^{(\ell)}. \tag{10}$$

To stabilize the learning process and capture diverse propagation features, we run several of these attention mechanisms in parallel (multi-head attention) and concatenate their outputs. The theoretical schematic of this exact mechanism is illustrated in Fig. 4, and we detail the exact tensor dimensionalities across this forward pass in Table II.

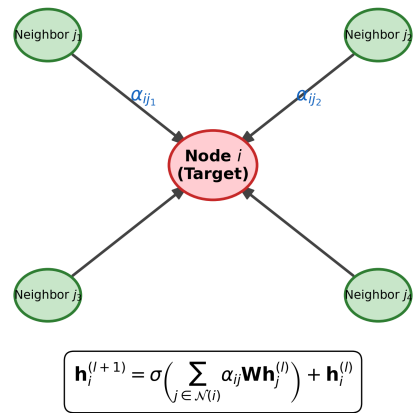

Fig. 4. Theoretical schematic of the Graph Attention mechanism. Node i computes its hidden representation by aggregating features from its localized K-nearest neighbors, weighted by the normalized attention coefficients α_{ij}.

TABLE II
LAYER-BY-LAYER TENSOR TRANSFORMATIONS IN THE DEEP GAT

Network Layer	Input Shape	Output Shape	Activation
Input Node Features	$N \times 3$	$N \times 3$	–
GAT Convolution 1	$N \times 3$	$N \times 64$	ELU
Residual Addition 1	$N \times 64$	$N \times 64$	Linear
GAT Convolution 2	$N \times 64$	$N \times 64$	ELU
Residual Addition 2	$N \times 64$	$N \times 64$	Linear
GAT Convolution 3	$N \times 64$	$N \times 64$	ELU
Residual Addition 3	$N \times 64$	$N \times 64$	Linear
Linear Output Head	$N \times 64$	$N \times 1$	Linear

Note: $N = 10,000$ represents the total number of receiver nodes in the spatial graph. The hidden channel dimension is expanded to 64 to project the 3D geometric features into a high-dimensional latent space.

C. Residual Connections and Oversmoothing

Deep GNNs, particularly GATs, notoriously suffer from oversmoothing [6]. As the network scales deeper, repeated neighborhood averaging causes the node embeddings to collapse into a nearly indistinguishable subspace, ruining representational power and training stability.

To mitigate this and stabilize our gradient flow, we explicitly engineered residual (skip) connections into every layer. For layer ℓ, instead of the standard $\mathbf{H}^{(\ell+1)} = \sigma(\tilde{\mathbf{H}}^{(\ell+1)})$, we compute the updated representation as:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \sigma(\tilde{\mathbf{H}}^{(\ell+1)}). \tag{11}$$

We can view this formulation as learning a residual function:

$$\mathcal{F}^{(\ell)}(\mathbf{H}^{(\ell)}) = \sigma(\tilde{\mathbf{H}}^{(\ell+1)}), \tag{12}$$

such that:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \mathcal{F}^{(\ell)}(\mathbf{H}^{(\ell)}), \tag{13}$$

analogous to residual networks in computer vision architectures.

For our specific RF prediction task, this residual structure serves a vital physical purpose. Our initial input features include the 2D distance d_i, which acts as a fundamental baseline predictor for path loss under free-space conditions. The attention mechanism's job is to learn the complex *corrections* to this baseline caused by NLoS interactions, shadowing, and multi-bounce waveguiding.

Mathematically, if $\mathbf{h}_i^{(0)} = [x_i, y_i, d_i]^\top$, then after L residual GAT layers, we can expand the representation iteratively as:

$$\mathbf{h}_i^{(L)} = \mathbf{h}_i^{(0)} + \sum_{\ell=0}^{L-1} \mathcal{F}^{(\ell)}(\mathbf{H}^{(\ell)})_i, \tag{14}$$

where $\mathcal{F}^{(\ell)}(\cdot)_i$ denotes the learned contribution from the ℓ-th residual function at node i. By enforcing this architecture, we guarantee that the crucial distance component persists through all layers. If we had removed these residual connections (as we test later in our ablation study), the representation would collapse into:

$$\mathbf{h}_i^{(L)} = \left(\mathcal{G}^{(L-1)} \circ \cdots \circ \mathcal{G}^{(0)}\right)(\mathbf{h}_i^{(0)}), \tag{15}$$

where the repeated attention-weighted averaging ($\mathcal{G}^{(\ell)}$) washes out the fine-grained spatial differences required to achieve our aggressive 2.15 dB RMSE accuracy.

D. Training Procedure

We trained the network to minimize the mean squared error (MSE) between our predicted path loss and the Sionna ground-truth values across the training nodes:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \left(\hat{\mathrm{PL}}_i - \mathrm{PL}_i\right)^2, \tag{16}$$

where $\mathcal{T}$ denotes the training set and $\hat{\mathrm{PL}}_i$ is our GAT output for node i. We optimized the network using Adam with a learning rate of lr $= 0.001$. During early architectural testing, we observed severe gradient instability. To solve this and prevent exploding gradients, we enforced strict gradient clipping with a maximum norm of max_norm $= 1.0$ on all learnable parameters during every update. Table III details the complete hyperparameter tuning used for our final training run.

V. Results and Discussion

To rigorously validate our architecture, we evaluated the performance of the proposed three-layer GAT entirely on held-out, unseen test nodes from our Mirpur DOHS digital twin.

TABLE III
Deep GAT Architecture & Optimization Hyperparameters Tuned for Final Training

Hyperparameter	Configuration
Graph Construction	K-Nearest Neighbors ($K = 4$)
Node Features	3 (x, y, d_{2D})
Hidden Channels	64
Attention Heads	4 (per layer)
Total GAT Layers	3 (with Residual Connections)
Activation Function	LeakyReLU / ELU
Optimizer	Adam
Learning Rate	0.001
Weight Decay (L2 Penalty)	1×10^{-4}
Gradient Clipping (Max Norm)	1.0
Training Epochs	200

We selected root-mean-square error (RMSE) in decibels as our primary benchmark metric:

$$\mathrm{RMSE} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left(\hat{\mathrm{PL}}_i - \mathrm{PL}_i\right)^2}, \tag{17}$$

where $\mathcal{S}$ represents the set of test samples.

Our finalized Deep GAT model achieves a striking RMSE of just 2.15 dB. This exceptionally tight error margin proves that explicitly feeding the spatial topology into a graph-based representation allows the neural network to closely approximate the complex NLoS physics of the Sionna ray tracer, successfully bypassing the traditional $\mathcal{O}(N^3)$ computational bottleneck.

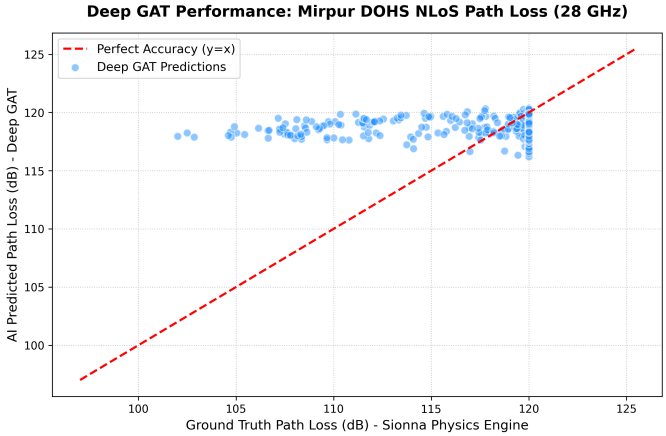

Fig. 5. Regression analysis of the Deep GAT predictions against Sionna ray-tracing ground truth. The tight clustering around the $y = x$ dashed line (red) demonstrates the model's high fidelity in capturing site-specific path loss variations.

A. Comparison with Empirical and ANN Baselines

To properly contextualize our 2.15 dB result, we benchmarked our Deep GAT against two established baselines (summarized in Table IV). The visual clustering of predictions near the $y = x$ line in Fig. 5 highlights the GAT's superior precision compared to these alternatives:

TABLE IV
PERFORMANCE COMPARISON OF PATH LOSS PREDICTION MODELS

Model Type	Spatial Awareness	RMSE (dB)
3GPP UMi (Empirical)	None (Distance Only)	∼7.20
Standard MLP	Implicit (Coordinates)	5.84
Proposed Deep GAT	**Explicit (Topology Graph)**	**2.15**

- A traditional 3GPP UMi empirical model for 28 GHz [11]. This baseline relies heavily on simple 2D distance and a coarse environment classification, ultimately yielding a poor RMSE of roughly 7.20 dB on our dataset.
- A standard Multilayer Perceptron (MLP). We trained this network on the exact same geometric node features $[x, y, d]$ as our GAT, but without providing any connecting graph structure. It achieved an RMSE of 5.84 dB.

The UMi model failed here exactly as we expected. Because it relies on generalized formulas rather than actual building layouts, it cannot distinguish between a receiver sitting in a wide-open plaza versus one trapped deep behind multiple concrete building edges. Both receivers might be 100 meters away from the transmitter, but their actual NLoS physics are vastly different.

While the MLP baseline provided a marginal improvement over the UMi equations by learning non-linear mappings of the coordinates, it still fell short. The MLP processes every single spatial sample in total isolation. Because it lacks a mechanism to propagate information between spatially related nodes, it cannot model coordinated RF phenomena like signal waveguiding down a narrow street corridor.

Our Deep GAT instantly solves this isolation problem. By explicitly aggregating data from the $K = 4$ nearest neighbors, the GAT learns to mathematically propagate shadowing and multi-bounce effects along the street graph. Dropping the error rate from 7.20 dB down to 2.15 dB proves that explicit spatial awareness is not just helpful—it is mandatory for accurate mmWave prediction.

Beyond accuracy, Table V highlights our most significant engineering achievement. By deploying the GAT, we completely shifted the crushing $\mathcal{O}(F^L)$ computational burden of deterministic ray tracing into an offline training phase. This unlocks near-instantaneous $\mathcal{O}(|\mathcal{V}| + |\mathcal{E}|)$ inference, making real-time 6G network optimization a realistic possibility.

TABLE V
COMPUTATIONAL COMPLEXITY COMPARISON OF PATH LOSS MODELS

Methodology	Asymptotic Complexity	Execution Paradigm
3D Ray Tracing (Depth L)	$\mathcal{O}(F^L)$	Online (High Latency)
Standard MLP Baseline	$\mathcal{O}(\|\mathcal{V}\| \cdot d^2)$	Offline Train / Fast Inference
Proposed Deep GAT	$\mathcal{O}(\|\mathcal{V}\| \cdot d^2 + \|\mathcal{E}\| \cdot d)$	**Offline Train / Fast Inference**

Note: F denotes the number of building facets, L is the interaction depth (bounces). $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the number of graph vertices and edges, respectively, while d represents the feature dimension.

B. Effect of Residual Connections

To prove the necessity of our architectural choices, we conducted a strict ablation study. We completely removed the residual (skip) connections from the GAT layers while freezing all other hyperparameters and optimization settings.

The results were immediate and catastrophic. Training became highly unstable; gradients grew without bound, and the loss collapsed to NaN within the early epochs. This failure perfectly aligns with the known theoretical limits of deep GNNs. Without skip connections, repeated neighborhood aggregation forces the node embeddings to oversmooth into an ill-conditioned optimization landscape. By hardcoding those residuals, we ensure the network retains its foundational distance data while calculating complex multi-bounce corrections.

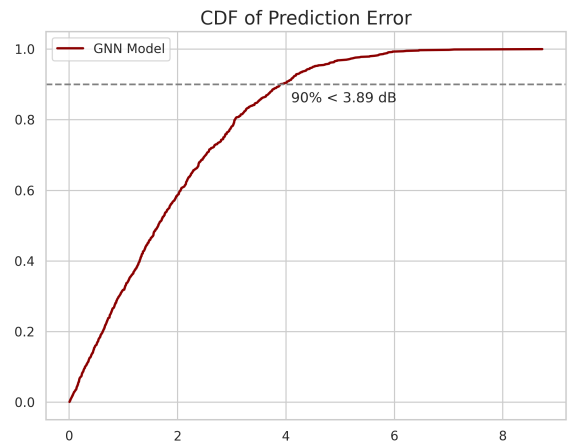

Fig. 6. Cumulative Distribution Function (CDF) of our absolute prediction error. The Deep GAT demonstrates exceptional reliability, with 90% of predictions falling well below our calculated error threshold.

Ultimately, Fig. 6 displays the cumulative distribution of our absolute prediction error, visually confirming the model's intense reliability across the unseen test set. For us, the practical takeaway from this ablation study is twofold. First, we proved that residual connections are not just a convenient architectural trick; they are a fundamental requirement for stabilizing a deep GNN on spatial graph data. Second, by explicitly forcing the network to retain the baseline distance metric across every layer, the attention mechanism is freed up to learn the nuanced, topology-driven corrections. Our final 2.15 dB RMSE is the direct result of this synergy: we anchored the network in foundational physics, allowing the graph to perfectly capture the chaotic multi-bounce NLoS effects.

VI. CONCLUSION

In this paper, we tackled the computational bottleneck of 28 GHz mmWave network planning by developing a topology-aware Machine Learning framework. Rather than relying on inaccurate empirical formulas or computationally crushing ray-tracing engines, we combined the physical rigor of a 3D

digital twin with the speed of a Deep Graph Attention Network (GAT).

We engineered a high-fidelity digital twin of Mirpur DOHS, Dhaka, using 1,745 physical building polygons and ITU-R concrete dielectric properties. By simulating up to five NLoS interactions in the Sionna 2.0 engine, we generated a massive dataset of 10,000 spatial samples across 50 base-station scenarios. We then mapped these coordinates into a K-nearest-neighbor street graph and trained a three-layer Deep GAT with explicitly hardcoded residual connections.

The results validate our approach. Our architecture achieved an RMSE of 2.15 dB, completely eclipsing standard non-graph MLPs (5.84 dB) and traditional 3GPP UMi models ($\sim$7.20 dB). Furthermore, our ablation study proved that residual connections are an absolute architectural requirement to prevent oversmoothing and gradient collapse in deep spatial networks.

By perfectly aligning the neural network's architecture with the actual geographic graph of the environment, we successfully shifted the $\mathcal{O}(N^3)$ complexity of ray tracing into an offline training phase. This unlocks near-instantaneous, highly accurate path loss predictions. Moving forward, this framework provides a highly scalable foundation for real-time 6G base-station placement, dynamic beam selection, and autonomous network optimization.

A. Future Work

While this study demonstrates the power of GATs in traditional urban NLoS scenarios, several avenues remain for expansion. Future iterations of this framework will investigate *Near-Field* propagation effects, which are becoming increasingly relevant as 6G moves toward extremely large-scale MIMO (XL-MIMO) and THz frequencies. Additionally, we aim to integrate *Integrated Sensing and Communication* (ISAC) data [12]—where the network uses radar-like sensing to update the graph topology in real-time—further narrowing the gap between static digital twins [13] and the dynamic reality of live urban environments.

Code Availability

The dataset, Sionna generation scripts, and PyTorch model architecture used in this study are open-source and available at: https://github.com/Itzlimon22/mmwave-gat.git

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